

# ANALYSTLAB AFRICA

## HealthConnect Clinic Experience Lab — Week 5

### Data Analytics Track — Initial HealthConnect Analytics Report

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## 1. Review of Week 4 Foundation

Week 4 established the problem: understanding what drives HealthConnect's patient appointment no-shows, using the Appointment Dataset (5,000 records) and Data Dictionary. Five business questions and five candidate KPIs were proposed, and an initial data quality pass confirmed the dataset was clean (no duplicates, no logical inconsistencies), with only minor gaps in reminder\_channel, distance\_to\_clinic\_km and waiting\_time\_minutes. The proposed Week 5 focus was to deepen the exploratory analysis, calculate the proposed KPIs, and produce initial visualisations — which this report delivers.

## 2. Data Preparation

Building on the Week 4 assessment, the cleaned dataset (HealthConnect\_Appointment\_Data\_Cleaned.csv) was used as the basis for this stage's analysis. No further structural changes were required; the original raw file remains untouched. A brief recap of the preparation applied:

- booking\_date and appointment\_date converted from text to proper date type
- reminder\_channel blanks (1,366 rows, 27.3%) filled with “None” — confirmed to correspond exactly to rows where reminder\_sent = No, so this reflects the absence of a reminder rather than a data error
- distance\_to\_clinic\_km (1.8% missing) and waiting\_time\_minutes (1.2% missing) left as-is; excluded row-wise only in analyses that specifically use these fields
- No duplicate appointment\_id or full-row duplicates found; no logical inconsistencies (negative lead times, appointment dates preceding booking dates, or previous\_no\_shows exceeding previous\_appointments)

## 3. Exploratory Data Analysis

The overall no-show rate across all 5,000 appointments is 48.5% (2,423 No-Show, 2,314 Attended, 263 Cancelled). No-show rate was examined against every relevant variable category:

| Dimension                                 | Range Found           | Signal Strength                                                        |
|-------------------------------------------|-----------------------|------------------------------------------------------------------------|
| Prior no-show history (0 to 3+ prior)     | 43.5% → 68.8%         | Strongest — 25 point spread                                            |
| Booking lead time (Same-day to 8+ days)   | 24.7% → 51.5%         | Very strong — 27 point spread                                          |
| Distance to clinic (0-5km to 20+km)       | 46.5% → 57.8%         | Moderate — 11 point spread, driven mainly by the 20+km group           |
| Reminder sent (Yes vs No)                 | 47.4% vs 51.4%        | Modest — 4 point difference                                            |
| Reminder channel (SMS / WhatsApp / Email) | 45.8% / 49.8% / 48.4% | Weak — SMS marginally best, not conclusive                             |
| Appointment type (4 categories)           | 46.6% – 51.2%         | Weak — Follow-up highest, General Consultation lowest                  |
| Age group / Gender                        | 45.1% – 50.7%         | Weak — no strong demographic pattern; only 65+ notably lower           |
| Waiting time                              | 47–49% across bands   | Not usable — the 60+min band contains only 3 records and is unreliable |

Cancellations (5.3% of all appointments, 263 records) were treated as a separate outcome category throughout rather than merged with No-Show, since they represent a materially different patient behaviour (proactive notice vs. silent absence) and are not the focus of this analysis.

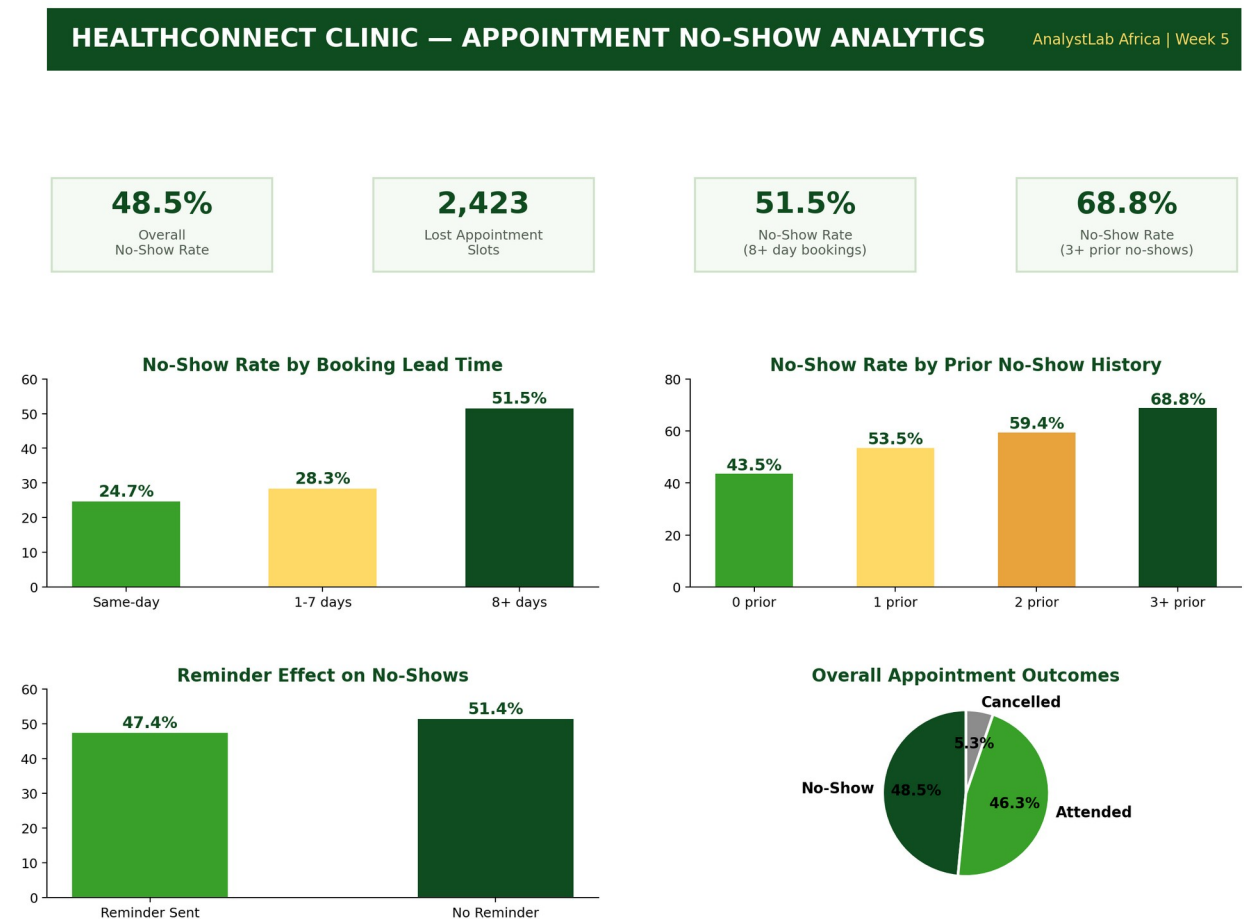
## 4. KPI Development

Five KPIs were selected from the Week 4 proposed list and calculated in full below.

| KPI                           | Business Question                                            | Result                | Interpretation                                                                           |
|-------------------------------|--------------------------------------------------------------|-----------------------|------------------------------------------------------------------------------------------|
| No-Show Rate (%)              | What proportion of appointments are missed?                  | 48.5%                 | Nearly 1 in 2 appointments is missed — confirms this is a major operational issue        |
| No-Show Rate by Lead Time     | Does booking lead time relate to attendance?                 | 24.7% / 28.3% / 51.5% | Longer lead times roughly double the risk of a no-show — a strong, actionable pattern    |
| No-Show Rate by Prior History | Which patients are most associated with missed appointments? | 43.5% → 68.8%         | The single strongest predictor found — past behaviour strongly predicts future behaviour |
| Reminder Effectiveness Rate   | Does receiving a reminder reduce no-shows?                   | 47.4% vs 51.4%        | A real but modest effect — reminders alone are not a complete solution                   |
| Lost Appointment Slots        | How much capacity is lost to no-shows?                       | 2,423 slots           | Nearly half of all scheduled capacity across the dataset went unused due to no-shows     |

## 5. Dashboard

An initial visual dashboard summarising the KPIs and key EDA findings is shown below.



## 6. Business Insights

- **Prior no-show history is the single strongest signal found: patients with 3+ previous no-shows miss appointments at a 68.8% rate — nearly 1.6x the overall average — making this the clearest candidate for a risk-flagging or targeted-intervention approach.**
- **Booking lead time shows a near-identical strength of pattern: same-day/within-week bookings have roughly half the no-show rate (24.7–28.3%) of appointments booked 8+ days ahead (51.5%). Encouraging shorter booking windows where clinically appropriate could meaningfully reduce no-shows.**
- **Reminders help, but modestly (4 percentage points) — not enough alone to solve the problem. Combined with the near-identical performance across reminder channels (45.8–49.8%), this suggests the reminder mechanism itself (timing, content, or delivery) may matter more than which channel is used, and is worth testing further rather than assumed effective as-is.**
- **Distance to clinic shows a moderate effect concentrated in the 20+km group (57.8%, vs ~46.5% for patients within 10km) — a relatively small but identifiable group that may benefit from telehealth or transport-support interventions.**
- **Demographics (age, gender) show no strong pattern, meaning interventions are more efficiently targeted by behaviour (prior history, lead time) than by who the patient is — an important scoping insight for any future prioritisation of resources.**

## 7. Cross-Track Collaboration

Track interacted with: Data Science.

Information exchanged: the KPI and EDA findings above — particularly that `previous_no_shows` and `booking_lead_days` are the two strongest signals found — are directly relevant to the Data Science track's choice of predictive features and target variable definition for their no-show prediction model.

Why it matters: aligning on which variables carry real signal (vs. weak ones like demographics) early avoids duplicated exploratory work between tracks and keeps the two tracks' conclusions consistent when the project reaches its Final Presentation stage.

Result: this analysis can be shared with the Data Science track as an input to their Week 5 feature selection, and their eventual model results will be used to validate (or refine) the strength of these findings in Week 6.

## 8. Assumptions, Limitations, Risks and Dependencies (Updated from Week 4)

### Still relevant from Week 4

- This is a fictional, synthetic dataset; findings are illustrative for the Experience Lab, not real clinical conclusions
- `distance_to_clinic_km` and `waiting_time_minutes` missingness (1.8% and 1.2%) is excluded row-wise rather than imputed

### Resolved or refined in Week 5

- The booking-lead-time relationship flagged in Week 4 for further validation is confirmed as a strong, consistent pattern with a large enough sample in every band (Same-day, 1-7 days, 8+ days) to be reliable

### New in Week 5

- The 60+ minute waiting-time band contains only 3 records; the apparent high no-show rate there is not statistically meaningful and should not be used to draw conclusions

- waiting\_time\_minutes is populated even for No-Show records, suggesting it reflects an expected/assigned wait time rather than an actually observed one — this limits its usefulness as a behavioural predictor and should be confirmed with the Data Dictionary owner if this analysis continues
- The 20+km distance band is a relatively small group (393 of 5,000 records); its 57.8% figure is directionally useful but should be treated with some caution given the smaller sample